BANGLADESH PHYSICS OLYMPIAD 2025

National Round

Category – C

Participant Name:

Registration Number: **Class:**

Name of Institution:

Contact No: **Email:**

Time: 2 Hour

READ THESE INSTRUCTIONS FIRST

- Make sure that Participant Name, Registration number, Class, Name of Institution and Contact number are written clearly.
- Answer all the questions. If there is discrepancy in the two versions use the data provided in the Bengali version.
- Scientific calculator is allowed.

Name of the Examiner:	**Score**
Signature:	

BdPhO-2024-2025, Category C

Feb 28, 2025

1 Houston, Do we have Gravitational Radiation?

An indirect evidence of gravitational radiation (hence of General relativity) came from the observations of the binary pulsar PSR 1913+16 by Hulse and Taylor, which was first recorded in 1975. This binary system consists of two neutron stars, orbiting around each other (about their center of mass), one of which is a pulsar (with a pulsar period 0.059 secs) who are . Pulsars are rapidly spinning neutron stars with a frequency in the range of 10-100 s^{-1}. It was found, during a period of more than 20 years of observation, that there is a very small decrease in the orbital period of this binary system.

Let us assume that the masses of the stars in the binary system are both equal to M and they execute a circular orbit of diameter D.

(a) If the period of the orbital motion in τ what is the tangential velocity v of each of the stars when the orbit diameter is D? **1**

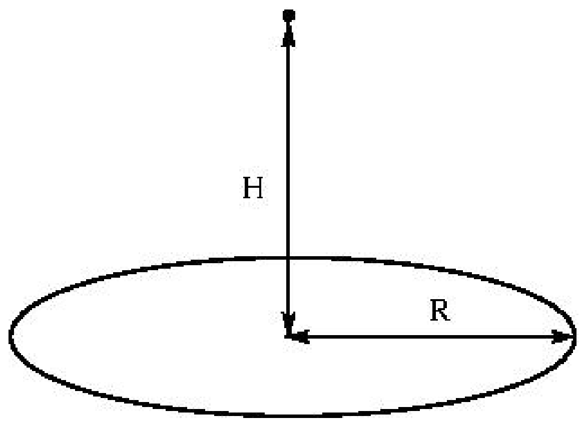
(b) What is the total energy of the system in terms of the velocity v and the orbit diameter? **1**

(c) In circular orbits the centripetal force is provided by the gravitational force. Using the expressions you obtained above, find how that total energy of the system varies with respect to the distance between the stars. **2**

(d) How does the time period of the orbit vary with respect to the total energy of the system? **1**

(e) If the system's energy changes with respect to time is $P = \frac{dE}{dt}$ (t beng time), what is the ratio of the rate of the change of orbit time period and the time period τ in terms of P and E? **3**

(f) The observations give us $\omega = 2.2 \times 10^{-4}$ and $M = 1.4 M_{sun}$. while the decrease of the orbit time was found to be $\frac{d\tau}{dt}$. Estimate the power radiated in the orbit. (Use $G = 6.674 \times 10^{-11}$ $m^3\ kg^{-1}\ s^{-2}$) **2**



2 Threading the Disc

A point mass of mass m and charge $-q$ ($q > 0$) is held at rest above the centre of a uniformly charged disk of radius R carrying charge $+Q$, with $Q > 0$ as well. (See the figure). Assume that there is a tiny hole at the centre of the disc so that the point mass can pass through it without any hinderance. The distance between the point mass and the center of the disk is given by H,

(a) Find the force on the point mass as a function of H. **2**.

(b) What happens to the the force when the disk radius $R \to \infty$ but keeping the charge per unit area on the disk fixed to σ? **1**

(c) If the point mass is released from the height H what is the velocity with which it passes through the center of the disc? **2**

(d) Find the frequency of the oscillation of the point mass for the motion in the vertical direction(through the disc) . **3**

3 Not the Fit Elegance

For a certain fluid, an experimentalist finds the amount of fluid flow V though a pipe is related to the pressure difference X between the pipe ends satisfies a power-law scaling relation: $V = AP^b$ where A and b are constants.

The following data points were obtained :

P	1.1	1.9	3.1	4.05	5.02
V	2.05	4.4	7.95	13.1	20.3

(a) How should one modify the original relation given above so that the fit can be done in terms of a straight line? **2**

(b)Plot the graph in the provided grid. Label the axes properly and ensure an appropriate scale. **2**

(c) What are the values of A and b that you get from your graph. **2**

(d) Can you Interpret the physical meaning of the exponent in this context? **1**

(Graph Grid provided in the next page)